C. Preventive measures

Background

The GDG considered the evidence and other relevant information to inform recommendations on antenatal interventions to prevent the following conditions.

- **Asymptomatic bacteriuria (ASB):** Defined as true bacteriuria in the absence of specific symptoms of acute urinary tract infection, ASB is common in pregnancy, with rates as high as 74% reported in some LMICs *(125)*. *Escherichia coli* is associated with up to 80% of isolates *(83)*. Other pathogens include *Klebsiella* species, *Proteus mirabilis* and group B streptococcus (GBS). While ASB in non-pregnant women is generally benign, in pregnant women obstruction to the flow of urine by the growing fetus and womb leads to stasis in the urinary tract and increases the likelihood of acute pyelonephritis. If untreated, up to 45% of pregnant women with ASB may develop this complication *(126)*, which is associated with an increased risk of preterm birth.

- **Recurrent urinary tract infections:** A recurrent urinary tract infection (RUTI) is a symptomatic infection of the urinary tract (bladder and kidneys) that follows the resolution of a previous urinary tract infection (UTI), generally after treatment. Definitions of RUTI vary and include two UTIs within the previous six months, or a history of one or more UTIs before or during pregnancy *(127)*. RUTIs are common in women who are pregnant and have been associated with adverse pregnancy outcomes including preterm birth and small-for-gestational-age newborns *(127)*. Pyelonephritis (infection of the kidneys) is estimated to occur in 2% of pregnancies, with a recurrence rate of up to 23% within the same pregnancy or soon after the birth *(128)*. Little is known about the best way to prevent RUTI in pregnancy.

- **Rhesus D alloimmunization:** Rhesus (Rh) negative mothers can develop Rh antibodies if they have an Rh-positive newborn, causing haemolytic disease of the newborn (HDN) in subsequent pregnancies. Administering anti-D immunoglobulin to Rh-negative women within 72 hours of giving birth to an Rh-positive baby is an effective way of preventing RhD alloimmunization and HDN *(129)*. However, Rhesus alloimmunization occurring in the third trimester due to occult transplacental haemorrhages will not be prevented by postpartum anti-D.

- **Soil-transmitted helminthiasis:** Over 50% of pregnant women in LMICs suffer from anaemia, and helminthiasis is a major contributory cause in endemic areas *(33)*. Soil-transmitted helminths are parasitic infections caused mainly by roundworms (*Ascaris lumbricoides*), hookworms (*Necator americanus* and *Ancylostoma duodenale*), and whipworms (*Trichuris trichiura*). These worms (particularly hookworms) feed on blood and cause further bleeding by releasing anticoagulant compounds, thereby causing iron-deficiency anaemia *(130)*. They may also reduce the absorption of iron and other nutrients by causing anorexia, vomiting and diarrhoea *(131)*.

- **Neonatal tetanus:** Tetanus is an acute disease caused by an exotoxin produced by *Clostridium tetani*. Neonatal infection usually occurs through the exposure of the unhealed umbilical cord stump to tetanus spores, which are universally present in soil, and newborns need to have received maternal antibodies via the placenta to be protected at birth. Neonatal disease usually presents within the first two weeks of life and involves generalized rigidity and painful muscle spasms, which in the absence of medical treatment leads to death in most cases *(132)*. Global vaccination programmes have reduced the global burden of neonatal tetanus deaths and continue to do so; estimates show a reduction from an estimated 146 000 in 2000 to 58 000 (CI: 20 000–276 000) in 2010 *(133)*. However, because tetanus spores are ubiquitous in the environment, eradication is not biologically feasible and high immunization coverage remains essential *(134)*.

In addition to GDG recommendations on the above, this section of the guideline includes two recommendations on disease prevention in pregnancy that have been integrated from WHO guidelines on malaria and HIV prevention that are relevant to routine ANC.

Women's values

A scoping review of what women want from ANC and what outcomes they value informed the ANC guideline *(13)*. Evidence showed that women from high-, medium- and low-resource settings valued having a positive pregnancy experience. This included the tailored (rather than routine) use of biomedical tests and effective preventive interventions to optimize pregnancy and newborn health, and the ability of health-care practitioners to explain and deliver these procedures in a knowledgeable, supportive and respectful manner (high confidence in the evidence).

C.1: Antibiotics for asymptomatic bacteriuria (ASB)

RECOMMENDATION C.1: A seven-day antibiotic regimen is recommended for all pregnant women with asymptomatic bacteriuria (ASB) to prevent persistent bacteriuria, preterm birth and low birth weight. *(Recommended)*

Remarks

- This recommendation should be considered alongside the recommendation on ASB diagnosis (Recommendation B.1.2).
- Stakeholders may wish to consider context-specific ASB screening and treatment based on ASB and preterm birth prevalence, as it may not be appropriate in settings with low prevalence.
- Evidence on preterm birth is of low certainty and large multicentre trials are needed to confirm whether screening and antibiotic treatment reduces preterm birth and perinatal mortality in LMICs. Such trials should also aim to evaluate the effects of group B streptococcus (GBS) screening and treatment.
- Studies have shown that GBS bacteriuria is a sign of heavy GBS colonization, which may not be eradicated by antibiotic treatment. GBS bacteriuria is a risk factor for having an infant with early onset GBS disease. WHO recommends that pregnant women with GBS colonization receive intrapartum antibiotic administration to prevent early neonatal GBS infection (see *WHO recommendations for prevention and treatment of maternal peripartum infections [135]*).
- Preterm birth indicators should be monitored with this intervention, as should changes in antimicrobial resistance.

Summary of evidence and considerations

Effects of antibiotics for ASB versus no antibiotics or placebo (EB Table C.1)

The evidence on the effects of antibiotics for ASB was derived from a Cochrane review that included 14 trials involving approximately 2000 women *(83)*. Most trials were conducted in HICs between 1960 and 1987. Types of antibiotics included sulfonamides, ampicillin, nitrofurantoin and some antibiotics that are no longer recommended for use in pregnancy, such as tetracycline. Treatment duration between trials varied widely from a single dose, to continuous treatment throughout pregnancy. Bacteriuria was usually defined as at least one clean-catch, midstream or catheterized urine specimen with more than 100 000 bacteria/mL on culture, but other definitions were also used.

Maternal outcomes

The only maternal ANC guideline outcomes reported were infection outcomes. Low-certainty evidence suggests that antibiotics may reduce persistent bacteriuria (4 trials, 596 women; RR: 0.30, 95% CI: 0.18–0.53)*;* however, the evidence on the effect on pyelonephritis is very uncertain.

Fetal and neonatal outcomes

Low-certainty evidence suggests that antibiotics for ASB may reduce low-birth-weight neonates (8 trials, 1437 neonates; RR: 0.64, 95% CI: 0.45–0.93) and preterm birth (2 trials, 142 women; RR: 0.27, 95% CI: 0.11–0.62). No other ANC guideline outcomes were reported.

Additional considerations

- The GDG also evaluated evidence on treatment duration (single dose versus short-course

[4–7 days]) from a related Cochrane review that included 13 trials involving 1622 women *(136)*. Ten trials compared different durations of treatment with the same antibiotic, and the remaining three compared different durations of treatment with different drugs. A wide variety of antibiotics was used. The resulting pooled evidence on bacterial persistence (7 trials), recurrent ASB (8 trials) and pyelonephritis (2 trials) was judged as very uncertain. However, on sensitivity analysis including high-quality trials of amoxicillin and nitrofurantoin only, the high-certainty evidence indicates that bacterial persistence is reduced with a short course rather than a single dose (2 trials, 803 women; RR: 1.72, 95% CI: 1.27–2.33). High-certainty evidence from one large trial shows that a seven-day course of nitrofurantoin is more effective than a one-day treatment to reduce low birth weight (714 neonates; RR: 1.65, 95% CI: 1.06–2.57). Low-certainty evidence suggests that single-dose treatments may be associated with fewer side-effects (7 trials, 1460 women; RR: 0.70, 95% CI: 0.56–0.88). See Web supplement (EB Table C.1).

- The GDG also evaluated evidence on the test accuracy of urine Gram staining and dipstick testing (see Recommendation B.1.2 in section 3.B).

Values

See "Women's values" at the beginning of section 3.C: Background (p. 64).

Resources

Antibiotic costs vary. Amoxicillin and trimethoprim are much cheaper (potentially around US$ 1–2 for a week's supply) than nitrofurantoin, which can cost about US$ 7–10 for a week's supply of tablets *(137)*. Repeated urine testing to check for clearance of ASB has cost implications for laboratory and human resources, as well as for the affected women. The emergence of antimicrobial resistance is of concern and may limit the choice of antimicrobials *(125)*.

Equity

Preterm birth is the leading cause of neonatal death worldwide, with most deaths occurring in LMICs; therefore, preventing preterm birth among disadvantaged populations might help to address inequalities.

Acceptability

In LMICs, some women hold the belief that pregnancy is a healthy condition and may not accept the use of antibiotics in this context (particularly if they have no symptoms) unless they have experienced a previous pregnancy complication (high confidence in the evidence) *(22)*. Others view ANC as a source of knowledge, information and medical safety, and generally appreciate the interventions and advice they are offered (high confidence in the evidence). However, engagement may be limited if this type of intervention is not explained properly. In addition, where there are likely to be additional costs associated with treatment, women are less likely to engage (high confidence in the evidence).

Feasibility

A lack of resources in LMICs, both in terms of the availability of the medicines and testing, and the lack of suitably trained staff to provide relevant information and perform tests, may limit implementation (high confidence in the evidence) *(45)*.

C.2: Antibiotic prophylaxis to prevent recurrent urinary tract infections (RUTI)

RECOMMENDATION C.2: Antibiotic prophylaxis is only recommended to prevent recurrent urinary tract infections in pregnant women in the context of rigorous research. *(Context-specific recommendation – research)*

Remarks

- Further research is needed to determine the best strategies for preventing RUTI in pregnancy, including the effects of antibiotic prophylaxis on pregnancy-related outcomes and changes in antimicrobial resistance.

Summary of evidence and considerations

Effects of prophylactic antibiotics to prevent RUTI compared with no antibiotics (EB Table C.2)

The evidence on the effects of prophylactic antibiotics to prevent RUTI was derived from a Cochrane review in which only one trial in the USA involving 200 pregnant women contributed data *(127)*. Women admitted to hospital with pyelonephritis were randomized, after the acute phase, to prophylactic antibiotics (nitrofurantoin 50 mg three times daily) for the remainder of the pregnancy plus close surveillance (regular clinic visits and urine culture, with antibiotics on positive culture), or to close surveillance only.

Maternal outcomes

Evidence from this single study on the risk of recurrent pyelonephritis and RUTI with prophylactic antibiotics is very uncertain. No other maternal ANC guideline outcomes were reported in the study.

Fetal and neonatal outcomes

Evidence on the risk of low birth weight and preterm birth with prophylactic antibiotics is very uncertain. No other fetal and neonatal ANC guideline outcomes were reported in the study.

Additional considerations

- Antibiotic prophylaxis to prevent RUTI may lead to increased antimicrobial resistance and there is a lack of evidence on this potential consequence.

Values

See "Women's values" at the beginning of section 3.C: Background (p. 64).

Resources

Antibiotic costs vary. Trimethoprim is cheaper than nitrofurantoin, which can cost about US$ 5 for 28 × 100 mg tablets *(137)*.

Equity

Impact not known.

Acceptability

In LMICs, some women hold the belief that pregnancy is a healthy condition and may not accept the use of antibiotics in this context (particularly if they have no symptoms) unless they have experienced a previous pregnancy complication (high confidence in the evidence) *(22)*. Others view ANC as a source of knowledge, information and medical safety and generally appreciate the interventions and advice they are offered (high confidence in the evidence). However, engagement may be limited if this type of intervention is not explained properly. In addition, where there are likely to be additional costs associated with treatment, women are less likely to engage (high confidence in the evidence).

Feasibility

A lack of resources in LMICs, both in terms of the availability of the medicines and testing, and the lack of suitably trained staff to provide relevant information and perform tests, may limit implementation (high confidence in the evidence) *(45)*.

C.3: Antenatal anti-D immunoglobulin prophylaxis

RECOMMENDATION C.3: Antenatal prophylaxis with anti-D immunoglobulin in non-sensitized Rh-negative pregnant women at 28 and 34 weeks of gestation to prevent RhD alloimmunization is recommended only in the context of rigorous research. *(Context-specific recommendation – research)*

Remarks

- This context-specific recommendation relates to anti-D prophylaxis during pregnancy and not the practice of giving anti-D after childbirth, for which there is high-certainty evidence of its effect of reducing RhD alloimmunization in subsequent pregnancies *(129)*. Anti-D should still be given postnatally when indicated.
- Determining the prevalence of RhD alloimmunization and associated poor outcomes among women in LMIC settings, as well as developing strategies to manage this condition, is considered a research priority

Summary of evidence and considerations

Effects of antenatal anti-D immunoglobulin prophylaxis in non-sensitized Rh-negative pregnant women compared with no intervention (EB Table C.3)

The evidence on the effects of antenatal anti-D prophylaxis was derived from a Cochrane review that included two RCTs involving over 4500 Rh-negative pregnant women *(138)*. Most participants were primigravidas. Both trials compared antenatal anti-D prophylaxis with no antenatal anti-D prophylaxis. One trial used a dose of 500 IU, the other used 250 IU, given at 28 and 34 weeks of gestation. Data were available for 3902 pregnancies, and more than half the participants gave birth to Rh-positive newborns (2297). All women with Rh-positive newborns received postpartum anti-D immunoglobulin as per usual management. The primary outcome was the presence of Rh-antibodies in maternal blood (a proxy for neonatal morbidity). No maternal ANC guideline outcomes (including maternal satisfaction and side-effects) and few perinatal guideline outcomes were reported in these trials.

Fetal and neonatal outcomes

Evidence on the effect of antenatal anti-D on RhD alloimmunization during pregnancy, suggesting little or no difference in effect, is very uncertain. In addition, the evidence on the effect on postpartum RhD alloimmunization and alloimmunization up to 12 months postpartum among women giving birth to Rh-positive newborns (n = 2297 and 2048, respectively) is very uncertain, partly because events were rare. Evidence on the effect of antenatal anti-D on neonatal morbidity (jaundice) from one trial (1882 neonates) is also very uncertain, partly because events were rare. No other ANC guideline outcomes were reported in the review.

Additional considerations

- Low-certainty evidence from the Cochrane review suggests that Rh-negative women who receive antenatal anti-D are less likely to register a positive Kleihauer test (which detects fetal cells in maternal blood) during pregnancy (1 trial, 1884 women; RR: 0.60, 95% CI: 0.41–0.88) and at the birth of a Rh-positive neonate (1 trial, 1189 women; RR: 0.60, 95% CI: 0.46–0.79).
- In the Cochrane review, the rate of RhD alloimmunization during pregnancy, the postpartum period and up to 12 months later among women in the control group was 0.6%, 1.1% and 1.5%, respectively.
- Rates of RhD alloimmunization in subsequent pregnancies were not reported in the trials.
- There is no evidence on optimal dose of antenatal anti-D prophylaxis and various regimens are used. There are two ongoing studies listed in the Cochrane review, which may help to clarify issues around effects and dosage once completed.
- Only 60% of Rh-negative primigravidas will have an Rh-positive newborn, therefore 40% of Rh-negative women will receive anti-D unnecessarily with antenatal anti-D prophylaxis *(138)*.

Values

See “Women’s values” at the beginning of section 3.C: Background (p. 64).

Resources

A single dose of anti-D can cost around US$ 50 (500 IU) to US$ 87 (1500 IU) *(139)*, depending on the brand and local taxes; therefore, the cost of antenatal prophylaxis for two 500 IU doses could be as much as US$ 100 per woman. Additional costs will include screening for blood typing in settings where Rh blood tests are not currently performed.

Equity

The contribution of RhD alloimmunization to perinatal morbidity and mortality in various LMIC settings is uncertain and it is not known whether antenatal anti-D for non-sensitized Rh-negative women will impact on equity.

Acceptability

Anti-D immunoglobulin is derived from human plasma and is administered by injection, which may not be acceptable to all women. Qualitative evidence indicates that engagement may be limited if tests and procedures are not explained properly to women, or when women feel their beliefs, traditions and social support mechanisms are overlooked or ignored by health-care professionals (high confidence in the evidence) *(22)*.

Feasibility

In a number of LMIC settings providers feel that a lack of resources, both in terms of the availability of the medicines and the lack of suitably trained staff to provide relevant information, may limit implementation of recommended interventions (high confidence in the evidence) *(45)*. Anti-D needs refrigeration at 2–8°C, which may not be feasible in some LMIC settings.

C.4: Preventive anthelminthic treatment

RECOMMENDATION C.4: In endemic areas,[a] preventive anthelminthic treatment is recommended for pregnant women after the first trimester as part of worm infection reduction programmes. *(Context-specific recommendation)*

Remarks

- This recommendation is consistent with the WHO *Guideline: preventive chemotherapy to control soil-transmitted helminth infections in high-risk groups (140)*, which states that:

  > "Preventive chemotherapy (deworming), using single-dose albendazole (400 mg) or mebendazole (500 mg) is recommended as a public health intervention for pregnant women, after the first trimester, living in areas where both: (1) the baseline prevalence of hookworm and/or *T. trichiura* infection is 20% or more and (2) where anaemia is a severe public health problem, with prevalence of 40% or higher among pregnant women, in order to reduce the burden of hookworm and *T. trichiura* infection (conditional recommendation, moderate quality of evidence)."

- Endemic areas are areas where the prevalence of hookworm and/or whipworm infection is 20% or more. Anaemia is considered a severe public health problem when the prevalence among pregnant women is 40% or higher.
- Infected pregnant women in non-endemic areas should receive anthelminthic treatment in the second or third trimester on a case-by-case basis *(140)*. A single dose of albendazole (400 mg) or mebendazole (500 mg) should be used *(140, 141)*.
- The safety of these drugs in pregnancy has not been unequivocally established; however, the benefits are considered to outweigh the disadvantages *(141, 142)*.
- WHO recommends a treatment strategy comprising two treatments per year in high-risk settings with a prevalence of ≥ 50% for soil-transmitted helminthiasis, and once per year in areas with a 20–50% prevalence *(140)*.
- For further guidance on soil-transmitted helminth infections, refer to the WHO *Guideline: preventive chemotherapy to control soil-transmitted helminth infections in high-risk groups* (currently in press) *(140)*.

a Greater than 20% prevalence of infection with any soil-transmitted helminths.

Summary of evidence and considerations

Effects of prophylactic anthelminthic treatment against soil-transmitted helminths administered in the second trimester of pregnancy compared with no intervention or placebo (EB Table C.4)

The following evidence on the effects of prophylactic anthelminthic treatment was derived from a Cochrane review that included four trials conducted in Peru, Sierra Leone and Uganda, involving 4265 pregnant women *(142)*. In two trials (Peru and Sierra Leone), the anthelminthic medication (albendazole or mebendazole) was administered as a single dose in the second trimester, with or without daily iron and folic acid supplements, irrespective of the presence of proven helminthiasis. The frequency of anaemia (Hb < 110 g/L) in these two trials was 56% and 47%, respectively, and the frequency of intestinal worms ranged from 20% to 64.2% for roundworm, 46.4% to 65.6% for hookworm, and 74.4% to 82% for whipworm. One small Ugandan trial administered a single dose of albendazole (400 mg) or placebo to women in the second trimester, irrespective of the proven presence of helminthiasis; baseline prevalence was 15%, 38% and 6% for ascariasis, hookworm and trichuriasis, respectively. The other Ugandan RCT contributed data on albendazole plus ivermectin versus ivermectin only, administered as single doses to pregnant women in the second trimester; all women were infected with an intestinal helminth at trial entry.

Maternal outcomes

Low-certainty evidence suggests that a single dose of albendazole or mebendazole in the second trimester of pregnancy may have little or no effect on maternal anaemia (defined as Hb < 11 g/dL) (4 trials, 3266 women; RR: 0.94; 95% CI: 0.81–1.10).

Fetal and neonatal outcomes

Moderate-certainty evidence indicates that a single dose of albendazole or mebendazole in the second trimester of pregnancy probably has little or no effect on preterm birth (2 trials, 1318 women; RR: 0.88, 95% CI: 0.43–1.78) or perinatal mortality (2 trials, 3385 women; RR: 1.09, 95% CI: 0.71–1.67). No other ANC guideline outcomes were reported in the review.

Additional considerations

- None of the trials in the Cochrane review evaluated effects of more than one dose of anthelminthics. Findings from large non-randomized studies (NRSs) suggest that prophylactic anthelminthic treatment may have beneficial effects for mothers and newborns living in endemic areas *(143–145)*:
  - One NRS, including approximately 5000 pregnant women in Nepal with a 74% prevalence of hookworm infection, reported a 41% reduction in six-month infant mortality among women receiving two doses of albendazole (one each in the second and third trimesters) compared with no treatment (95% CI: 18–57%) *(143)*. This study also showed reductions in severe maternal anaemia with albendazole.
  - A study from Sri Lanka involving approximately 7000 women compared mebendazole with no treatment and found fewer stillbirths and perinatal deaths among women receiving mebendazole (1.9% vs 3.3%; OR: 0.55, 95% CI: 0.40–0.77), and little difference in the occurrence of congenital anomalies (1.8% vs 1.5%, for intervention and controls, respectively; OR: 1.24, 95% CI: 0.80–1.91), even among the 407 women who had taken mebendazole in the first trimester against medical advice *(145)*.
- The WHO manual on *Preventive chemotherapy in human helminthiasis* stresses that every opportunity should be taken to reach at-risk populations through existing channels *(141)*.
- Cross-referencing other WHO guidelines, the upcoming 2016 WHO *Guideline: preventive chemotherapy to control soil-transmitted helminth infections in high-risk groups* recommends that a single dose of albendazole or mebendazole should be offered to pregnant women in the second and third trimesters of pregnancy where the prevalence of any soil-transmitted helminth infection (roundworm, hookworm and whipworm) exceeds 20% *(140)*.
- Preventive helminthic treatment helps to lessen the burden of other infections, e.g. HIV, malaria and TB, and contributes to a sustained reduction of transmission *(142)*.

Values

See "Women's values" at the beginning of section 3.C: Background (p. 64).

Resources

Preventive chemotherapy against helminthic infections is a cost–effective intervention. The market price of a single tablet of generic albendazole (400 mg) or mebendazole (500 mg) is about US$ 0.02–0.03 *(141)*.

Equity

Helminthic infections are widely prevalent in poverty-stricken regions and control of this disease aims to alleviate suffering, reduce poverty and support equity *(141)*.

Acceptability

Affected women are often asymptomatic and may not perceive the need for treatment. Therefore, the prevalence of soil-based helminthiasis in a particular setting is likely to influence women's and providers' preferences. Studies of anthelminthic programmes among non-pregnant cohorts, e.g. schoolchildren, in endemic areas have shown high levels of acceptability *(146)*. For women receiving preventive treatment in endemic areas, worms are often visible in the stools the day after treatment, and this may reinforce the value of the intervention. However, where there are likely to be additional costs associated with treatment (high confidence in the evidence) or where the intervention is unavailable because of resource constraints (low confidence in the evidence) women may be less likely to engage with services *(45)*.

Feasibility

In a number of LMIC settings providers feel that a lack of resources, both in terms of the availability of the medicines and the lack of suitably trained staff to provide relevant information, may limit implementation of recommended interventions (high confidence in the evidence) *(45)*.

C.5: Tetanus toxoid vaccination

RECOMMENDATION C.5: Tetanus toxoid vaccination is recommended for all pregnant women, depending on previous tetanus vaccination exposure, to prevent neonatal mortality from tetanus. *(Recommended)*

Remarks

- This recommendation is consistent with recommendations from the 2006 WHO guideline on *Maternal immunization against tetanus (134)*. The GDG endorses the 2006 guideline approach, which recommends the following.
  - If a pregnant woman has not previously been vaccinated, or if her immunization status is unknown, she should receive two doses of a tetanus toxoid-containing vaccine (TT-CV) one month apart with the second dose given at least two weeks before delivery. Two doses protect against tetanus infection for 1–3 years in most people. A third dose is recommended six months after the second dose, which should extend protection to at least five years.
  - Two further doses for women who are first vaccinated against tetanus during pregnancy should be given after the third dose, in the two subsequent years or during two subsequent pregnancies.
  - If a woman has had 1–4 doses of a TT-CV in the past, she should receive one dose of a TT-CV during each subsequent pregnancy to a total of five doses (five doses protects throughout the childbearing years).
- Tetanus vaccination and clean delivery practices are major components of the strategy to eradicate maternal and neonatal tetanus globally *(147)*.
- Effective surveillance is critical for identifying areas or populations at high risk of neonatal tetanus and for monitoring the impact of interventions.
- A monitoring system should include an immunization register, personal vaccination cards and maternal health records, which should be held by the woman.
- For effective implementation, ANC health-care providers need to be trained in tetanus vaccination and the vaccine, equipment and supplies (refrigerator, needles and syringes) need to be readily available at ANC services.
- Policy-makers in low prevalence/high-income settings may choose not to include tetanus vaccination among ANC interventions if effective tetanus immunization programmes and good post-exposure prophylaxis exist outside of pregnancy.
- ANC contacts should be used to verify the vaccination status of pregnant women, and administer any vaccines that are recommended in the national immunization schedule. ANC contacts are also opportunities to explain the importance of infant vaccination and communicate the infant/child vaccination schedule to pregnant women.
- Further information can be found in the WHO guidance *(134)*, available at: http://www.who.int/reproductivehealth/publications/maternal_perinatal_health/immunization_tetanus.pdf; and in WHO's vaccine position papers, available at: http://www.who.int/immunization/documents/positionpapers/en

Summary of evidence and considerations

Effects of antenatal tetanus toxoid (TT) vaccination compared with no, other or placebo vaccination (EB Table C.5)

The evidence on the effects of TT vaccination was derived from a Cochrane review that assessed the effect of tetanus vaccination in women of reproductive age or pregnant women to prevent neonatal tetanus *(148)*. Two RCTs contributed data: one was conducted in Colombia between 1961 and 1965 and compared a tetanus vaccine (aluminium phosphate adsorbed tetanus toxoid [10LF]; 3 doses) with an influenza vaccine (1618 women, 1182 neonates); the other was conducted in the USA and compared a combined vaccine (tetanus/diphtheria/acellular pertussis [Tdap]; 1 dose) with saline placebo in 48 pregnant women between 30 and 32 weeks of gestation. Due to the relative paucity of RCT data, additional evidence on effects is also considered in the "Additional considerations" section.

Maternal outcomes
Low-certainty evidence suggests that local side-effects, such as pain, were more common with the Tdap vaccination than placebo (48 women; RR: 3.94, 95% CI: 1.41–11.01). There is no evidence on other maternal outcomes.

Fetal and neonatal outcomes
Low-certainty evidence from the Colombian trial suggests that there may be fewer neonatal tetanus cases among neonates whose mothers receive TT vaccination than among those who do not (1182 neonates; RR: 0.20, 95% CI: 0.10–0.40). Moderate-certainty evidence suggests that two or more doses of TT probably reduce neonatal mortality from any cause (1 trial, 688 neonates; RR: 0.31, 95% CI: 0.17–0.55). Further low-certainty evidence suggests that neonatal mortality from tetanus may be reduced among neonates whose mothers receive at least two TT doses (1 trial, 688 neonates; RR: 0.02, 95% CI: 0.00–0.30), but not among neonates whose mothers receive only one dose (1 trial, 494 neonates; RR: 0.57, 95% CI: 0.26–1.24). Congenital anomalies and other ANC guideline outcomes were not reported in the trials.

Additional considerations
- A systematic review that pooled data from the Colombian trial with that of a large cohort study of antenatal TT vaccination from India found moderate-certainty evidence to support a large effect (94% reduction) on neonatal tetanus deaths in favour of TT vaccination with at least two doses in pregnant women and women of childbearing age (2 trials, 2146 neonates; RR: 0.06, 95% CI: 0.02–0.20) *(149)*.
- TT vaccination has been widely used over 40 years, leading to a substantial decrease in neonatal tetanus and an increase in neonatal survival, with no sign of possible harm to pregnant women or their fetuses *(150)*. The WHO strategy for eliminating maternal and neonatal tetanus includes immunization of pregnant women, supplementary immunization activities in selected high-risk areas, promotion of clean deliveries and clean cord practices, and reliable neonatal tetanus surveillance *(134)*.

Values
See “Women’s values” at the beginning of section 3.C: Background (p. 64).

Resources
The cost of three doses of TT vaccine has been estimated at around US$ 3 per woman *(151)*, although lower costs in vaccination programmes have been reported *(152)*. The need for cold-chain equipment and staff training may add to costs.

Equity
Most deaths from neonatal tetanus occur in countries with low coverage of facility-based births, ANC and tetanus vaccination *(149)*. In addition, in LMICs, ANC coverage and infant mortality is often unequal between the most- and least-educated, urban and rural, and richest and poorest populations *(29)*. Therefore, increasing tetanus immunity in LMICs and among disadvantaged populations could help to address inequalities.

Acceptability
Qualitative evidence indicates that most women view ANC as a source of knowledge, information and medical safety, and generally appreciate the interventions and advice they are offered. However, engagement may be limited if vaccinations are not explained properly or when women feel their beliefs, traditions and social support mechanisms are overlooked or ignored by health-care professionals (high confidence in the evidence) *(22)*. Lack of engagement may be compounded if services are delivered in a hurried, inflexible, didactic manner (high confidence in the evidence).

Feasibility
Antenatal services provide a convenient opportunity for vaccinating pregnant women, particularly in settings without effective childhood immunization programmes. Qualitative evidence indicates that if there are additional costs associated with vaccination (including transport costs and loss of earnings), uptake may be limited (high confidence in the evidence) *(22)*. In addition, ANC providers in many LMIC settings feel that a lack of resources, both in terms of the availability of vaccines and the lack of suitably trained staff, may limit implementation (high confidence in the evidence) *(45)*.

C.6: Intermittent preventive treatment of malaria in pregnancy (IPTp)

RECOMMENDATION C.6: In malaria-endemic areas in Africa, intermittent preventive treatment with sulfadoxine-pyrimethamine (IPTp-SP) is recommended for all pregnant women. Dosing should start in the second trimester, and doses should be given at least one month apart, with the objective of ensuring that at least three doses are received. *(Context-specific recommendation)*

Remarks

- This recommendation has been integrated from the WHO *Guidelines for the treatment of malaria* (2015), where it is considered to be a strong recommendation based on high-quality evidence *(153)*.
- Malaria infection during pregnancy is a major public health problem, with substantial risks for the mother, her fetus and the newborn. WHO recommends a package of interventions for preventing and controlling malaria during pregnancy, which includes promotion and use of insecticide-treated nets, appropriate case management with prompt, effective treatment, and, in areas with moderate to high transmission of *Plasmodium falciparum*, administration of IPTp-SP *(153)*.
- The high-quality evidence supporting this recommendation was derived from a systematic review of seven RCTs conducted in malaria-endemic countries, which shows that three or more doses of sulfadoxine-pyrimethamine (SP) is associated with reduced maternal parasitaemia, fewer low-birth-weight infants and increased mean birth weight compared with two doses only *(154)*.
- The malaria GDG noted that most evidence was derived from women in their first and second pregnancies; however, the limited evidence on IPTp-SP from women in their third and subsequent pregnancies was consistent with benefit *(153)*.
- To ensure that pregnant women in endemic areas start IPTp-SP as early as possible in the second trimester, policy-makers should ensure health system contact with women at 13 weeks of gestation. Policy-makers could also consider supplying women with their first SP dose at the first ANC visit with instructions about the date (corresponding to 13 weeks of gestation) on which the medicine should be taken.
- SP acts by interfering with folic acid synthesis in the malaria parasite, thereby inhibiting its life-cycle. There is some evidence that high doses of supplemented folic acid (i.e. 5 mg daily or more) may interfere with the efficacy of SP in pregnancy *(155)*. Countries should ensure that they procure and distribute folic acid supplements for antenatal use at the recommended antenatal dosage (i.e. 0.4 mg daily).
- The malaria GDG noted that there is insufficient evidence on the safety, efficacy and pharmacokinetics of most antimalarial agents in pregnancy, particularly during the first trimester *(153)*.
- Detailed evidence and guidance related to the recommendation can be found in the 2015 guidelines *(153)*, available at: http://www.who.int/malaria/publications/atoz/9789241549127/en/

C.7: Pre-exposure prophylaxis for HIV prevention

RECOMMENDATION C.7: Oral pre-exposure prophylaxis (PrEP) containing tenofovir disoproxil fumarate (TDF) should be offered as an additional prevention choice for pregnant women at substantial risk of HIV infection as part of combination prevention approaches.
(Context-specific recommendation)

Remarks

- This recommendation has been integrated from the *WHO guideline on when to start antiretroviral therapy and on pre-exposure prophylaxis for HIV* (2015), where it is considered to be a strong recommendation based on high-quality evidence *(99)*. The evidence and further guidance related to the recommendation can be found in this guideline.
- "Substantial risk" is provisionally defined as HIV incidence greater than 3 per 100 person-years in the absence of PrEP, but individual risk varies within this group depending on individual behaviour and the characteristics of sexual partners. Local epidemiological evidence concerning risk factors and HIV incidence should be used to inform implementation.
- Thresholds for offering PrEP may vary depending on a variety of considerations, including resources, feasibility and demand.
- The level of protection is strongly correlated with adherence.
- Detailed evidence and guidance related to this recommendation can be found in the 2015 guideline *(99)*, available at: http://www.who.int/hiv/pub/guidelines/earlyrelease-arv/en/